SmolVLA on LIBERO

Faster, more efficient inference, and a voice-to-action pipeline

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Lightning talk · NVIDIA Jetson AI Lab Research Group · 10 November 2026

1
Faster inference

2
Original vs Tuned

3
Voice to action

Setup

HARDWARE	**Laptop GPU**: RTX 5090 (Blackwell), 24 GB	· not a Jetson
ROBOT	**Franka Panda arm**, simulated in LIBERO	· MuJoCo physics
POLICY	**SmolVLA**, run with LeRobot	· Original and Tuned
VOICE	**Whisper large-v3-turbo** + **Qwen3.5-4B**	· speech → command

Everything runs on this one GPU.

Three pillars, measured together

A speed-up only counts if accuracy holds and energy drops.

SPEED

111.0 ms

per decision

ACCURACY

82%

328 of 400 episodes succeed

EFFICIENCY

11.0 J

GPU energy per decision

= GPU energy counter ÷ decisions

The starting point: **Original** SmolVLA, with LeRobot's default settings.

400 episodes = 40 LIBERO tasks × 10 starting scenes each

Measured 8 Nov 2026

Where it started: an open pull request

GPU	Cameras / Prefix tokens	Full `predict_action_chunk` (Eager -> Compiled)	10-step Euler Denoise Loop (Eager -> Compiled)
RTX 4090	1 cam (113 tokens)	**207.2 ms -> 42.3 ms (4.9x)**	**184.0 ms -> 19.0 ms (9.7x)**
NVIDIA H200 (@arun-elr)	3 cams (198 tokens)	**179.4 ms -> 42.6 ms (4.2x)**	**149.6 ms -> 13.4 ms (11.2x)**
RTX 5090 Laptop (@captof-omuamua)	3 cams (240 tokens)	**100.3 ms -> 45.3 ms (2.2x)**	— (not timed separately)

My laptop, the loop alone: 75–79 → 18–19 ms, about 4×.

Measured 7 Nov, not posted yet.

PR #4634 by NotNANtoN (Anton)

LeRobot · still open

SmolVLA reads the scene once, then refines its action in 10 steps.

The PR keeps that reading fixed (the KV cache), so the 10 steps can be compiled.

RTX 4090	4.9×
H200	4.2×
My laptop	**2.2×**

I reviewed it and ran it. My numbers are in the table.

github.com/huggingface/lerobot/pull/4634 · screenshot of the PR description, 10 Nov 2026

From 2.2× to 7.7×: what I added

PR #4634 alone 10 steps, compiled loop



2.2× 100.3 → 45.3 ms

1 denoising step, batched cameras LeRobot setting + my port



4.0× 111.0 → 27.9 ms

Tuned: plus whole-model compile LeRobot's switch



7.7× 111.0 → 14.4 ms

Tuned = the same model and weights, with faster settings. Nothing was retrained.

These two settings also run on plain LeRobot today: 15.4 ms, 7.2×.

7.7× faster

7.6× less energy

11.0 → 1.44 J per decision

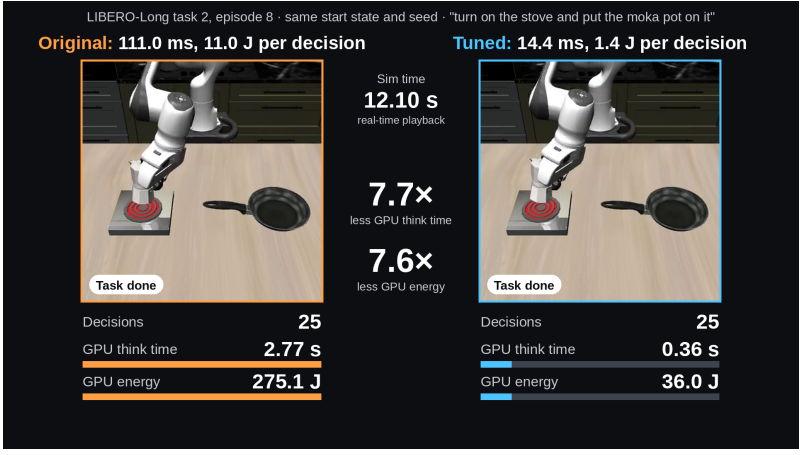
82% success, both

328 of 400 episodes each

The price: Tuned's first start takes about 2.5 minutes to compile. After that, 6.5 s.

Measured 8 Nov 2026, batch 1 · same 400 starting scenes for both · PR-alone row: 30 Oct · plain-LeRobot time: 10 Nov, speed only

▸ Demo 1: Original vs Tuned, same scene



The simulator waits for the model, so both arms move in step. **Watch the counters.**

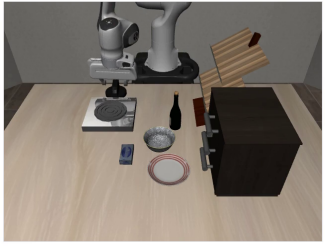
Counters = decisions so far × the measured time and energy per decision · measured 8 Nov 2026

Voice to action: the pipeline



A command	goes to SmolVLA as one of its 10 trained sentences
Two commands	run in order; the arm returns to its ready pose in between
Stop / continue	the arm holds, then resumes
Anything else	refused: the robot does nothing

▸ Demo 2a: one spoken command



MOVING 3.6 s

Heard: “Turn on the stove.”
Speech 55 ms + decision 0.2 ms

Command:
turn on the stove

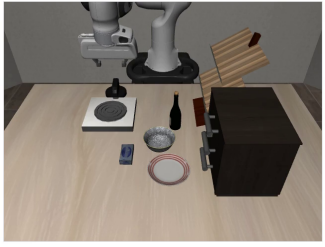
Recorded voice, played in at the marked times. Simulator.

3 of 3 runs
this command, on 3 starting scenes

0.2 s
from the last word to the first move

3.9 s
of movement to finish, in this run

▸ Demo 2b: stop and continue



Recorded voice, played in at the marked times. Simulator.

HOLDING 4.2 s

Heard: “Stop.”
Speech 46 ms + decision 0.0 ms

STOP

Command:
turn on the stove

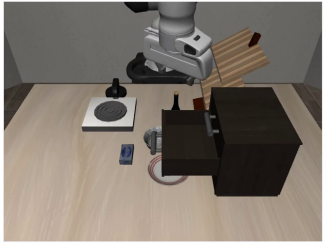
9 of 9 runs
finish after “stop”, then “continue”

5–10 mm
overrun after “stop”, then the arm holds

190 ms
from the end of the word to the hold

Recorded voice, simulator · 3 commands × 3 starting scenes · the arm then holds within about 5 mm · a live microphone would add time to detect the end of speech

▸ Demo 2c: two commands in one sentence



RETURNING 10.6 s

Heard: “Open the middle drawer, then turn on the stove.”
Speech 70 ms + decision 0.2 ms

Commands (2, in order):
1. open the middle drawer of the cabinet (done)
2. turn on the stove

Returning to the ready pose

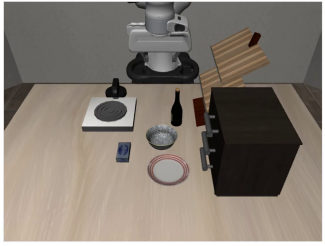
Recorded voice, played in at the marked times. Simulator.

5 of 5 runs

both done, for this sentence

The arm returns to its ready pose between the two commands.

▸ Demo 2d: a misheard word is refused



WAITING 5.0 s

Heard: “Kindly keep the ball on the plate.”

Speech 61 ms + decision 98.5 ms

Refused: not one of the robot's commands

Recorded voice, played in at the marked times. Simulator.

I said “bowl”.
It heard “ball”.

3 of 3 tries

refused: no command has a ball

0.35 mm

arm movement in the next 3 s

Thank you

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